On Orbit Determination

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Lambert's problem using universal variable approach

1. The Lambert's problem is concerned with the determination of orbit parameters from two position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ and the specified transfer time Δt [1, 2, 4].
2. Using a universal variable (or anamoly) χ and the Stumpff functions $C(z)$ amd $S(z)$, the Lagrange's f and g coefficients [3] are written as

$$f = 1 - \frac{\chi^2}{r_1} C(z) \qquad g = \Delta t - \frac{1}{\sqrt{\mu}} \chi^3 S(z) \tag{1}$$

$$\dot{f} = \frac{\sqrt{\mu}}{r_1 r_2} \chi [z S(z) - 1] \qquad \dot{g} = 1 - \frac{\chi^2}{r_2} C(z) \tag{2}$$

3. The Stumpff's functions are approximated using power series expansion and the value of z is determined using Newton's iterative method.

Lambert's problem using universal variable approach

4. From $f, g, \dot{f}, \dot{g}$ velocities can be computed as

$$\mathbf{v}_1 = \frac{1}{g}(\mathbf{r}_2 - f\mathbf{r}_1) \tag{3}$$

$$\mathbf{v}_2 = \frac{1}{g}(\dot{g}\mathbf{r}_2 - \mathbf{r}_1) \tag{4}$$

5. Using $\mathbf{r}_1$ (or $\mathbf{r}_2$) and $\mathbf{v}_1$ (or $\mathbf{v}_2$) we may obtain the orbital parameters using Kepler's equations.

Lambert's problem using universal variable approach

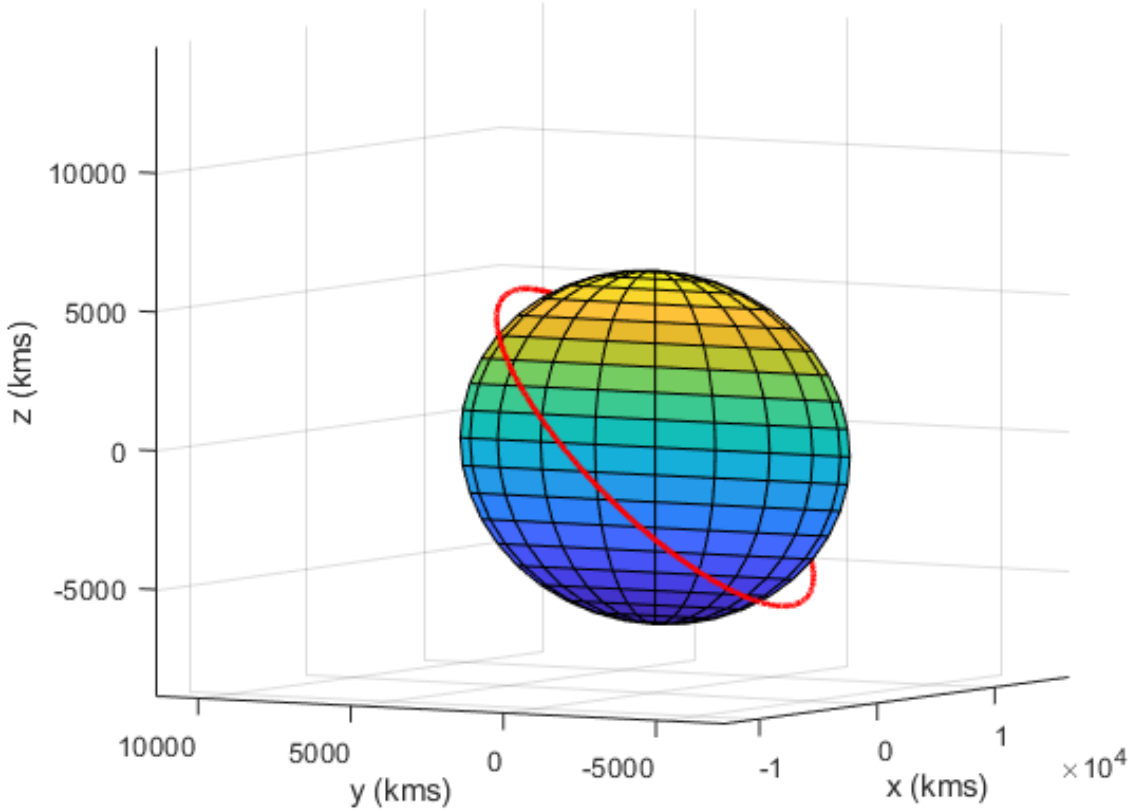


Figure: Orbit from estimated parameters.

Batch Least-Squares Estimation

1. Estimate initial position and velocity of a satellite using differential correction
2. position measurements
3. Equations:
   - 3.1 Measurement model

$$\mathbf{z}(t_i) = \mathbf{h}(t_i, \mathbf{x}_0) + \epsilon \tag{5}$$

   - 3.2 Cost function

$$J(\mathbf{x}_0) = (\mathbf{z} - \mathbf{h}(\mathbf{x}_0))^T (\mathbf{z} - \mathbf{h}(\mathbf{x}_0)) \tag{6}$$

   - 3.3 Least squares solution:

$$\mathbf{x}_0^{lsq}(i+1) = \mathbf{x}_0^{lsq}(i) + (\mathbf{H}^T \mathbf{W} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{W} \Delta \mathbf{z}(t_i) \tag{7}$$

where

- $\mathbf{z}$ are measurements at time t_i
- $\mathbf{x}_0$ is the initial state
- $\mathbf{h}$ is observation model
- $\mathbf{W}$ weight matrix ($= \text{diag}[\sigma_1^{-2}, \ldots, \sigma_n^{-2}]$)

Batch Least-Squares Estimation

The Jacobian $\mathbf{H}$ is a measure of model sensitivity to initial state

$$\mathbf{H} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}_0} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}} . \frac{\partial \mathbf{x}}{\partial \mathbf{x}_0} = \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \mathbf{\Phi}(t, t_0) \tag{8}$$

Since the observation model used here is $\mathbf{h} = \mathbf{r}$,

$$\frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \begin{bmatrix} \mathbf{I}_{3\times 3} & \mathbf{0}_{3\times 3} \end{bmatrix} \tag{9}$$

The state transition matrix $\mathbf{\Phi}(t, t_0)$ is a composition of Keplerian-to-cartesian and cartesian-to-Keplerian partial derivatives. For unperturbed analysis, the STM can be analytically computed by carefully evaluating the partial derivatives in

$$\Phi(t, t_0) = \left(\frac{\partial \mathbf{x}(t)}{\partial \boldsymbol{\alpha}(t)}\right) \mathbf{\Phi}_{\boldsymbol{\alpha}}(t, t_0) \left(\frac{\partial \mathbf{x}(t)}{\partial \boldsymbol{\alpha}(t)}\right)^{-1} \tag{10}$$

Batch Least-Squares Estimation

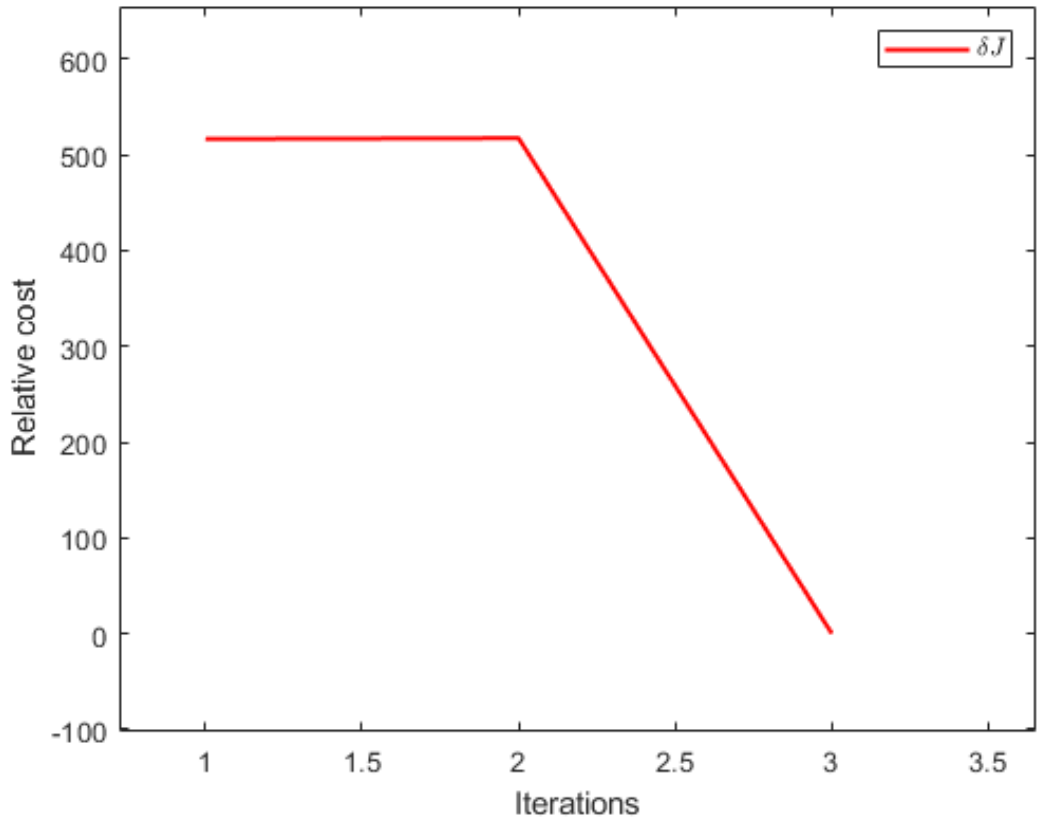


Figure: Relative cost convergence.

Simulation exercise: [a,e,i,Ω,ω,M] = [$R_{earth} + 650km$, 0.001, 51°,0,0,0]

Sequential Estimation

- Sequential orbit determination using extended Kalman filtering (EKF) - propagate and update steps.
- Start from an initial state estimate $\mathbf{x}_0$ and covariance matrix $\mathbf{P}_0$
- Propagation between measurement updates and state correction during update

Initialize	$\hat{\mathbf{x}}_0 = \hat{\mathbf{x}}(t_0)$ $\mathbf{P}_0 = \text{diag}[\sigma_x^2 \quad \sigma_y^2 \quad \sigma_z^2 \quad \sigma_{vx}^2 \quad \sigma_{Vz}^2 \quad \sigma_{vz}^2]$
Propagate	$\hat{\mathbf{x}}_i^- = \mathbf{\Phi}_i(\mathbf{x}_{i-1}^+)$ $\mathbf{P}_i^- = \mathbf{\Phi}_i \mathbf{P}_{i-1}^+ \mathbf{\Phi}_i^T$
Update	$\mathbf{K}_i = \mathbf{P}_i^- \mathbf{H}_i^T (\mathbf{W}_i^{-1} + \mathbf{H}_i \mathbf{P}_i^- \mathbf{H}_i^T)^{-1}$ $\mathbf{x}_i^+ = \mathbf{x}_i^- + \mathbf{K}_i(\mathbf{z}_i - \mathbf{h}_i(\mathbf{x}_i^-))$ $\mathbf{P}_i^+ = [I - \mathbf{K}_i \mathbf{H}_i)]\mathbf{P}_i^-$

EKF

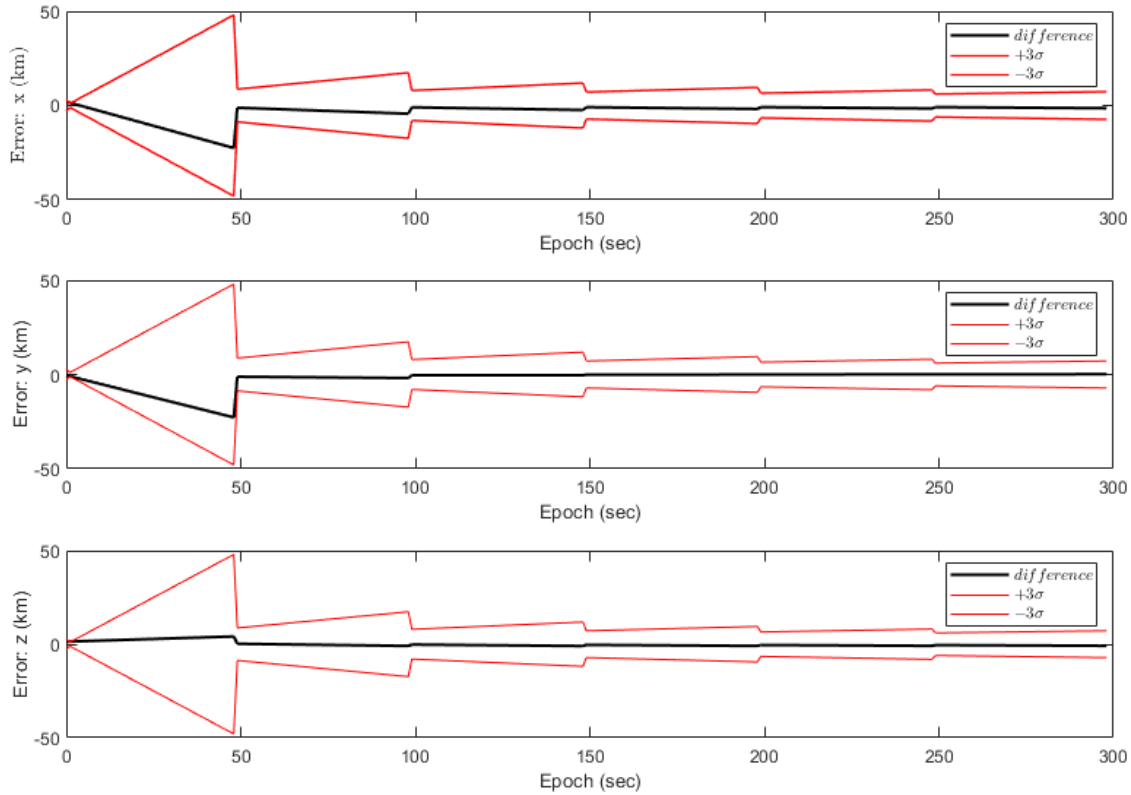


Figure: Estimation error in **r**.

EKF

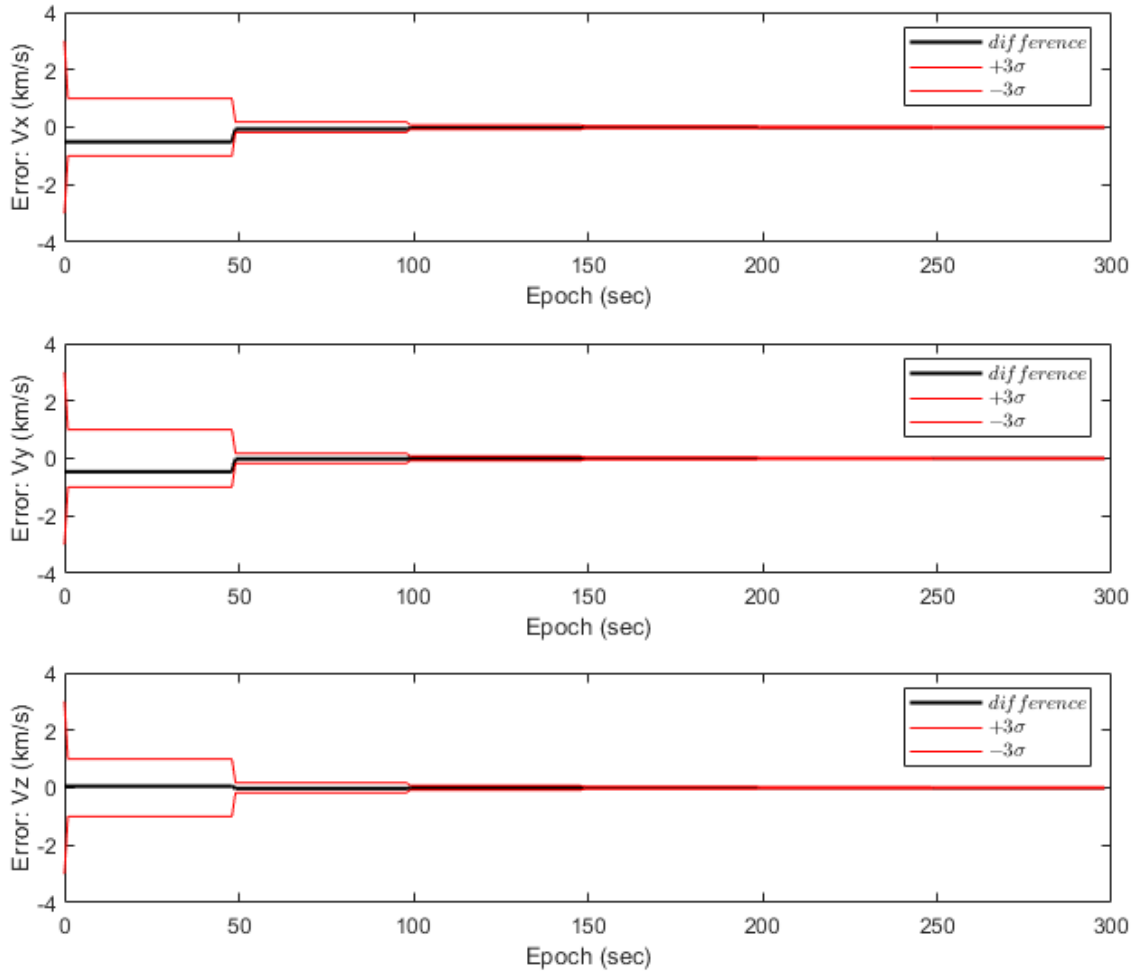


Figure: Estimation error in $\dot{\mathbf{r}}$.

Efforts and learnings

- Significant time during this project is spent on careful computation of the rigorous analytical form of the STM. Mistakes are penalized often via rank deficiency errors and ill-conditioning.
- Partials using equinoctial elements were tried for coordinate comparison but the involved mathematical derivation and software debugging discouraged me to complete those ideas toward this presentation.

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